THE MIDNIGHT MISSION

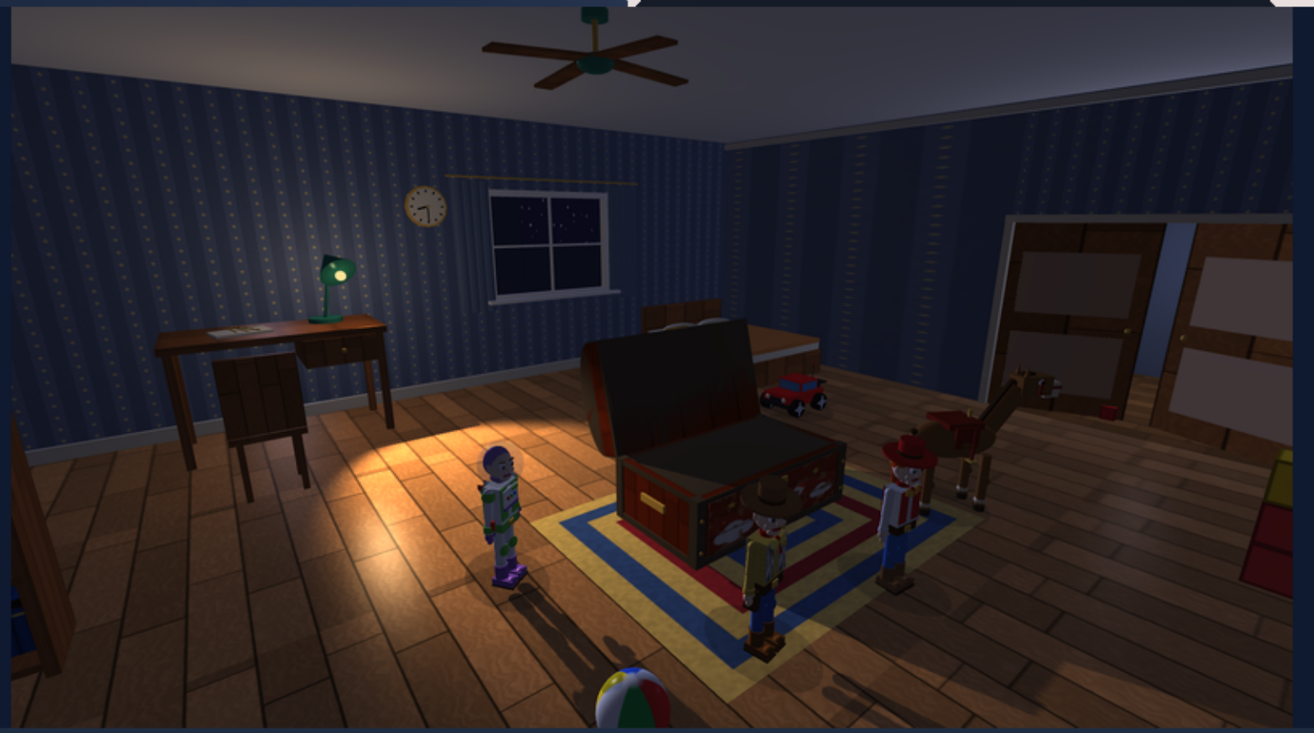
A haunted toy room. An interactive graphics story.

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CSE-4102 · Computer Graphics and Image Processing Laboratory

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ROOM NIGHT

EXPLORE, CONTROL, COMPARE

The proposal is realised as an eight-stage story from night to morning.

Play as Penny or any toy; the story layer drives the rest

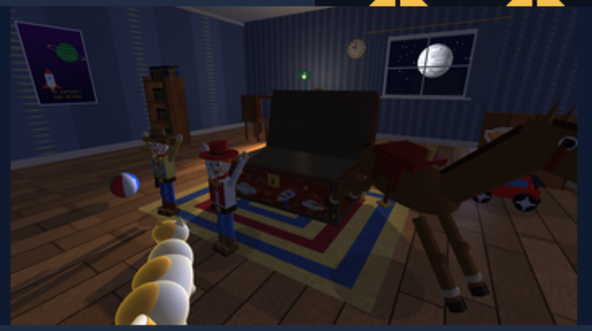
TRS + hierarchy; chest lid, mounting, Bullseye jump

Three light types + a sky dome outdoors

27 surface maps; analytic BVH ray tracing



HOUSE



TOYS ALIVE

PRIMITIVES BECOME CHARACTERS

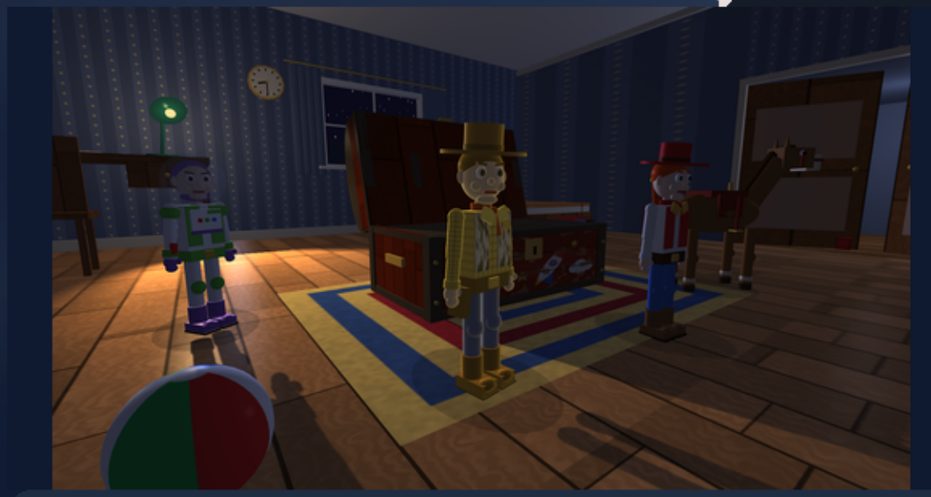
Position, normal and UV data define five indexed mesh families.

Plane 4 vertices / 2 triangles; cube 24 / 12

Sphere: 24 × 36 → 925 vertices / 1656 triangles

Cylinder: 134 / 128; cone: 99 / 64

The scene uses shared buffers and instance transforms



WOODY



BUZZ



PENNY

A WORLD WITH SURFACE DETAIL

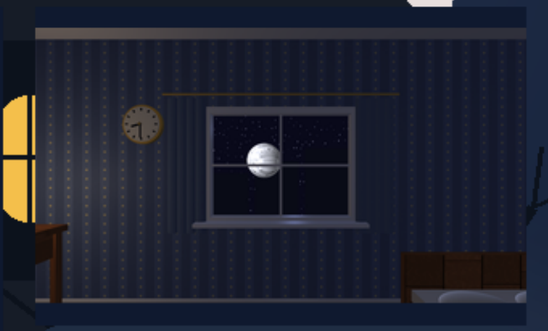
A connected house, garden, hallway, stairs and furnished room.

- Room furniture, curtains, shelves and wall art
- Garden, porch and the complete house exterior
- Connected floors, stairs and hinged doors

- Night-to-morning sky, lighting and shadows



HOUSE



WINDOW



BOOKCASE

COLOUR AND DETAIL FROM UVs

Surface maps add material detail while the primitive geometry stays shared.

Primitive UVs and per-material repeats

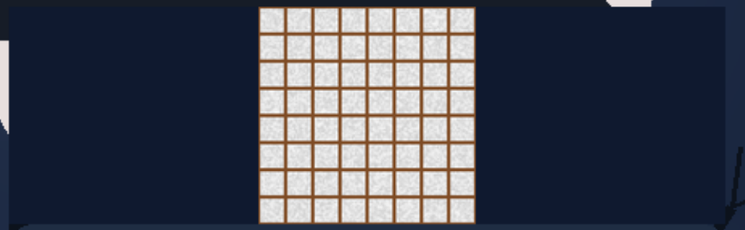
Wood, fabric, lunar craters and fence cutouts

Bilinear filtering and mipmaps

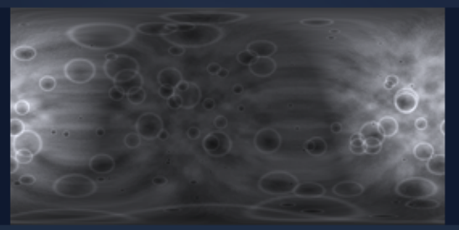
27 maps + white; the same array serves ray tracing



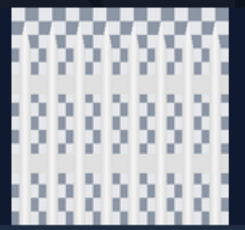
WOOD FLOOR



SHIRT PLAID



LUNAR SURFACE



PICKETS

IMPLEMENTATION EQUATIONS

$$(u', v') = (s_u u, s_v v), \qquad \mathbf{A} = \mathbf{k}_{RGB} \odot \text{texture}(u', v', \ell)_{RGB}$$

$$\alpha_{tex} < 0.5 \implies \text{discard}, \qquad N_{layers} = 27 + 1 = 28$$

ONE PARENT MOVES THE RIDER

World transforms propagate from parent to child.

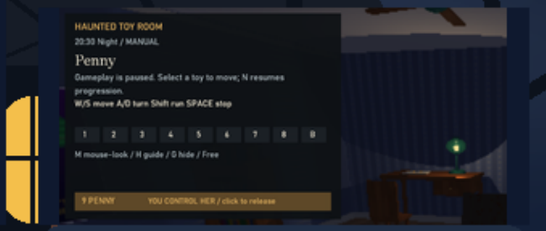
- Local translation, rotation and scale
- Mounting uses parent-child transforms
- Drive one actor while the others continue
- Inverse-transpose normals under scale and shear



MOUNTED



LIVE CONTROL



HUD PENNY PANEL

IMPLEMENTATION EQUATIONS

$$\mathbf{M}_w = \mathbf{M}_{parent}\,\mathbf{M}_{local}, \qquad \mathbf{M}_{local} = \mathbf{TRS}$$

$$\mathbf{n}_w = \text{normalize}\left((\mathbf{A}^{-1})^T \mathbf{n}_o\right)$$

EVERY MOTION HAS A RULE

Delta time connects displacement, gait, wheel rotation and camera motion.

Gait and joint animation follow travelled distance

Wheel and ball rotation follow arc length

Hinged chest, doors and wardrobe

Camera easing uses frame-independent response



RESCUE JUMP



BUZZ FLIGHT



CROWD AFTER

IMPLEMENTATION EQUATIONS

$$\mathbf{p}_{t+\Delta t} = \mathbf{p}_t + \mathbf{v}\Delta t, \qquad \Delta\theta = \frac{s}{r}$$

$$a = 1 - e^{-k\Delta t}, \qquad \mathbf{x}_{new} = (1-a)\mathbf{x}_{old} + a\mathbf{x}_{target}$$

THREE LIGHT TYPES

Ambient + diffuse + specular + emission

Directional sun/moon: attenuation = 1

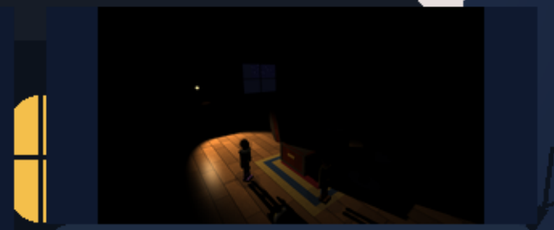
Point bulb: distance attenuation

Spotlight: smooth inner and outer cones

Ambient, diffuse, specular and emission



SKY DAY RASTER



SPOT



LAMP

IMPLEMENTATION EQUATIONS

$$\mathbf{C} = \mathbf{A} \odot k_a \mathbf{I}_a + \sum_i v_i a_i \mathbf{I}_i \odot \left(\mathbf{A} k_d [\mathbf{N} \cdot \mathbf{L}_i]_+ + k_s [\mathbf{R}_i \cdot \mathbf{V}]_+^n \right) + \mathbf{E}$$

$$a(d) = \frac{1}{k_c + k_l d + k_q d^2}, \qquad S = \text{smoothstep}(\cos\theta_{out}, \cos\theta_{in}, -\mathbf{L} \cdot \mathbf{D})$$

WHERE IS LIGHT EVALUATED?

Matched views expose the difference between vertex and pixel illumination.

Flat: one normal per triangle

Gouraud: interpolate vertex illumination

Phong: light the interpolated normal per pixel

Blinn: a half-vector specular highlight



FLAT



GOURAUD



PHONG



BLINN

IMPLEMENTATION EQUATIONS

$$\mathbf{C}_G = \sum_i \widehat{\lambda}_i \mathbf{C}_i, \qquad \mathbf{N}_P = \text{normalize}\left(\sum_i \widehat{\lambda}_i \mathbf{N}_i\right), \quad i \in \{0, 1, 2\}$$

$$\mathbf{H} = \text{normalize}(\mathbf{L} + \mathbf{V}), \qquad I_{s,\,Blinn} = k_s [\mathbf{N} \cdot \mathbf{H}]_+^{4n}$$

RAYS REVEAL A SECOND VIEW

Primary ray → nearest analytic hit → illumination → bounded continuation

BVH selects the nearest analytic hit

Shadow rays test light visibility

Two continuations by default; three surface hits

Mirror reflection and straight transmission



ROOM RAY



SKY NIGHT RT

IMPLEMENTATION EQUATIONS

$$\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}, \qquad \mathbf{d}_{reflect} = \mathbf{d} - 2(\mathbf{d} \cdot \mathbf{N})\mathbf{N}$$

$$\mathbf{C}_{acc} \leftarrow \mathbf{C}_{acc} + Tw\mathbf{C}_{local}, \qquad T \leftarrow T\rho, \quad T_0 = 1, \quad b = 2$$

THANK YOU

All five escape into the morning. The world stays open.

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CSE-4102

Geometry · hierarchy · motion · interaction

Lighting · shading · textures · ray tracing



STORY END

PROJECT DEMONSTRATION

The complete recorded walkthrough with the project's controls and graphics features.



Play in PowerPoint | Standalone: Presentation-Demo.mp4